Verifying the Menzerath-Altmann law on verbal constructions in 180 languages

Anonymous submission

Abstract

blabla

Keywords: xxx, yyy

1. Introduction

The current version of the Universal Dependencies (UD) collections of syntactic treebanks allows us to evaluate (a number of) universal properties on a set of 180 languages (De Marneffe et al., 2021). In this paper, we study the (typological) validity of the so-called Menzerath-Altman law (MAL) on the nature of complexity in linguistic units in terms of the relationship between the size of a given unit and the size of its components in any given domain (Menzerath, 1928, 1954; Altmann, 1980). Roughly speaking, the bigger the number of components of a given unit, the smaller their respective length. In the verbal domain, for instance, this implies that the length of verbal arguments decreases as the valency or number of verbal dependents increases (Mačutek et al., 2017). Our aim is also to propose a reliable metric to evaluate this hypothesis in the (unbalanced) UD collection with annotated corpora of significantly different sizes - including the so-called LOL languages, with millions of tokens, but also a number of corpora from less-studied languages with less than a thousand tokens. Furthermore, in order to be able to make typologically relevant claims based on an ad hoc language sample which is not construed for such purposes in the first place, we compare a number of typologically valid subsets of our sample to ensure that our conclusions are replicable beyond the sample as a whole and are not mere artefacts of sampling bias. Another important contribution of our study is that we looked at preverbal and postverbal domains separately. That is, we calculated two additional sets of metrics in addition to our general (or combined) MAL metrics, respectively LMAL and RMAL (for left/preverbal and right/postverbal arguments). Our results show that MAL is a clear typologically wide-spread preference across the languages of the world, but it is not an absolute universal principle. Importantly, we find that while MAL is a strong statistical cross-linguistic preference, there are not only languages in which this preference is trivial, but also languages which display the opposite tendency. Furthermore, our results show a clear difference between the pre- and postverbal domains. On the one hand, the MAL preference is stronger in the postverbal domain and, on the other hand, the anti MAL preference is stronger in the preverbal domain. Importantly, our findings show that in the postverbal domain VO languages are more likely to show a strong RMAL preference and inversely OV languages are more likely to show a strong LMAL preference in the preverbal domain.

In the following, we first briefly present MAL and the previous studies on which we are building (Section 2). We next present our data extraction based on the UD annotation (Section 3) and the different metrics we use in this study and motivate our methodological choices (Section 4). We then present our results and discuss our findings (Section 5).

2. The Menzerath-Altman law in the verbal domain

MAL predicts a negative correlation between constituents' average length and their total number in any given domain. MAL has been previously studied in the verbal domain using the UD annotation schema in Czech (Mačutek et al., 2017), as well as in the 2.3 version of the UD collection with 76 languages (Tanaka-Ishii, 2021). Our study builds on the latter while proposing a number of substantial improvements in order to be able to draw more reliable and typologically valid conclusions. First, we study MAL in the preverbal and postverbal domains sepertayly, in addition to the bilateral MAL. Second, our data extraction is more fine-grained than in Tanaka-Ishii's study (see Section 3). Third, we use the latest version of the UD, with 180 languages, offers a more important diversity of languages around the world. Moreover, we analyse the data while taking into account language families as well as language types, that is, VO, OV or no dominiant order (NDO). Finally, we discuss the choice of the best metric to use between different options.

3. Constituents in a the verbal construction in UD

In this paper, we suppose a certain knowledge of the UD annotation scheme. We just recall that the syntactic structure of all sentences is encoded by a dependency tree. Here we focus on verbal constructions, that is a verb and all the constituents that depend on it. Note that we decide to consider all lexical verbs and to not restrain our study to the main verbs (see (1b)) for at least two reasons: first it can double or triple the number of constructions and this is not insignificant when the corpus is not big; second, in some sentences, the most interesting verb, in terms of construction, can be subordinated because the main verb is only a modal verb for instance.

Not all units depending on a verb can be considered as constituents in the verbal construction. We thus eliminated punctuation signs (which in UD is attached by a relation `punct`), discourse markers (relation `discourse`) (see *alas* in (1a)), juxtaposed clauses or parentheses (relation `parataxis`), conjuncts in a coordination (relation `conj`) (1c), or coordinating conjunctions (relation `cc`) (see *but* in (1a)). We excluded the `vocative` relation, which concerns addresses to a participant (*Guys, take it easy!*), but kept dislocated arguments (*These guys, I don't trust them.*), especially because in some languages the line between governed units and dislocated units is difficult to draw. We also exclude the `aux` relation (see *do* in (1a)), considering that auxiliaries are part of the verb form, as well as compound (see *out* in 1d). The markers of the subordination such as subordinating conjunctions and adpositions (relations `mark` and `case`), which are analyzed as dependents in UD but are generally considered as heads of the constructions, are also discarded.

Some examples of verbal constructions in English as considered in this study are given in (1); the verb is underlined and the constituents are shown between square brackets; elements that are filtered are given in italics.

(1) a. *But* [I] , *alas* , *do* [not] <u>know</u> [how to see sheep] [through the walls of boxes]. en_littleprince

b. This means that *when* [you] <u>move</u> [a series or category field] [to the filter area and back], previously hidden items are again hidden. en_lines

c. [You] <u>tell</u> [me] [what they look like] *and I'll tell you what they are*. en_childes

d. [One minister] [reportedly] <u>handed</u> *out* [100 dollar 'gifts'] [to journalists attending a press conference for Allawi], [a practice that brings back bad memories to many Iraqis]. en_ewt

We also took into account the fact that a unit can be broken in two continuous parts in case of extraction or extraposition and decided to include these cases as two separate constituents. For instance, in (2a), the relative pronoun *what* is extracted and separated from its governor *do*, and, in (2b), the relative clause *that go well with the Santorum cocktail* is extraposed and separated from its governor *dishes*.

(2) a. [What] *do* [you] <u>like</u> [to do] ? en_ewt

b. [What culinary dishes] *would* [you] <u>recommend</u> [that go well with the Santorum cocktail]? en_gum

For each language L and each $n \geq 1$, we extracted all the verbal constructions with n constituents. We then calculated the size of each constituent, punctations excluded. In other words, we kept all the relations except `punct` and if the constituent contains a coordination or a discourse marker there is no reason to remove them. We call MAL$_n$(L) the mean size of the constituents in all the verbal constructions with n constituents. Note that we compute MAL$_n$(L) in a given language L only when we have at least 100 constructions with n constituents.

We define LMAL$_n$(L) and RMAL$_n$(L) in the same way. For LMAL, we select all the constructions with n preverbal constituents (i.e. on the left), regardless on the other size of the verb. Same thing for RMAL with all the constructions with n postverbal constituents (i.e. on the right).

4. Defining the MAL effect in the UD collection

We call λMAL(L) the function that mapst MAL$_n$(L) to each n. It has been shown by Altmann (1980) that a function such as λMAL(L) should follow a power law. In other words, MAL$_n$(L) is expected to be more or less proportional to $n^{-\beta}$ where β is a positive value (generally smaller than 1), which we call the MAL effect for L. Note that if MAL$_n$(L) $\approx n^{-\beta}$, then log(MAL$_n$(L)) $\approx -\beta$log(n). In other word, $-\beta$ is the slope of the regression line of the log-log representation of λMAL(L). The greater is β, the quicker is the decrease, the stronger is the MAL effect. Figure 1 gives the value of β (1.171) as well as the coefficient R^2 (0.892), which shows that λMAL(German) is close to a power law and that MAL is verified for German.

Tanaka-Ishii (2021) showed that, on her data, the values for n = 1 and n = 2 may not be aligned and she took onto account only the values for $n \geq 3$ for

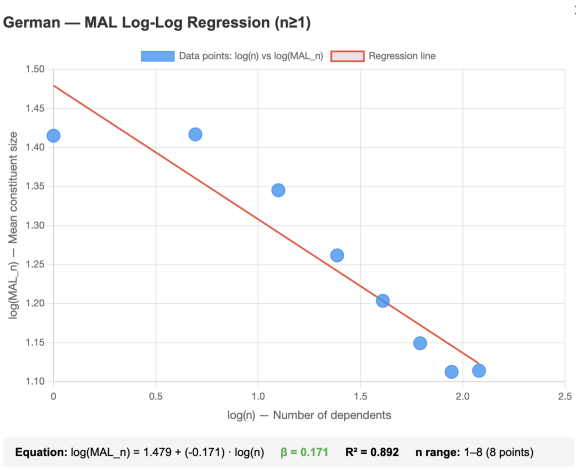

Figure 1: λMAL(German).

computing β. We think that the problem was was partly due to the fact that her data were not filtered.

We call $\beta(j \to k)$ the slope of λMAL for $j \leq n \leq k$. If $k = j + 1$, $\beta(j \to k)$ is just the slope between the point (log(MAL$_j$),log(j)) and (log(MAL$_k$),log(k)). In the figure 2, we compare the β(1$\to$2) with β(2$\to \infty$). The important dispersion suggests that the value for MAL$_1$ is effectively poorly aligned with the other values. The case where MAL$_1$ is aligned are the point on the diagonal $y = x$.

Nevertheless we observe that the values of MAL$_1$ can be sometimes under the regression line for $n \geq 2$ (as it is the case for German, Fig. 1 and sometimes above (see Czech, Fig. 4, or Arabic, Fig. 3).

We then decide to take β(1$\to \infty$) as the metric to evaluate the MAL effect. We also do that because, due to the filter and the fact that many treebanks are quite small, we do not have values of MAL$_n$ for $n \geq 4$. And it is even more problematic when we look at LMAL$_n$ and RMAL$_n$.

We nevertheless must mention an additional problem with n = 1. While the number of configurations with n constituents decrease for n $geq2$ only 20 of the 131 languages that have at least 100 configurations for n = 2 and 3 have more data n = 1 than for n = 2. Among them, 12 language have less that 100 configurations with one constituent (n = 1) and we must take β(2$\to \infty$) to evaluate the MAL effect.

There are 131 languages in our sample for which we have calculated a value for β(1$\to \infty$). We identify three types of behavior in our sample and givent that our values of β(1$\to \infty$) are mainly comprised between -0.3 and 0.3, we devide them into the three categories following categories:[1]

- Language L with β(1$\to \infty$) ≥ 0.1 is considered to have a MAL effect;
- Language L with β(1$\to \infty$) ≤ -0.1 is considered to have an Anti MAL effect.
- Language L with $-0.1 \leq \beta$(1$\to \infty$) ≤ 0.1 is considered to be in the grey zone;

The majority of languages of our sample (that is, 62 languages which make 47% of the 131 languages) fall into the first category, that is they show a MAL effect. But there are also a smaller number of languages (10 languages,),showing an Anti MAL effect. The remaining languages are in the gray zone (45% of languages).

One of the languages that do not follow a power law is OldEast Slavic, which not only has an Anti MAL effect (that is, the mean length of the constituents increase with the number of constituents), but on top of that the anti MAL effect increases with the number of constituents (Figure 5

5. Results

Out of the 186 languages available in our data, we caluclated a β(1$\to \infty$) for 131. In these sample we identify 62 languages with a MAL preference, 10 with an Anti MAL preference and 59 in the gray zone. Importantly, these three categories are fairly comparabely well-represented across languages families (Table 5) as well as main syntacticlanguge types, namely, head-final/OV, head-initial/VO, mixed/NDO. (Table 5). For language families, given the unbalanced nature of our sample, we have grouped the languages into the two main Indo-European (IE) and non-Indo-European (non-IE) groups.

For RMAL (postverbal domain) and LMAL (preverbal domain) we have calculted β(1$\to \infty$) for respectively 124 and 103 languages out of the total of 186 languages (Table **??** and Table **??**)). Interestingly, we find a clear difference between the pre- and postverbal domains. On the one hand, while we identify 62 languages showing a MAL effect, there are 81 languages (79%) which show a MAL effect in the postverbal domain (or a RMAL effect), against 39 languages (31%) showing a MAL effect in the preverbal domain (or a LMAL effect). In other words, 79% of languages show an MAL effect in the postverbal domain, against 31% in the preverbal domain, while the percentage of languages showing a combined MAL effect is 47%. On the other hand, while we identify 10 Anti MAL languages, there are 29 Anti LMAL languages and 7 Anti RMAL. In other words, 23% of languages show an anti MAL effect

[1] For MAL, no β(1$\to \infty$) are lesser than -0.3 and only 9 languages have a β(1$\to \infty$) greater than 0.3. Almost same thing for LMAL. For RMAL, 48 languages have a value greater than 0.3!

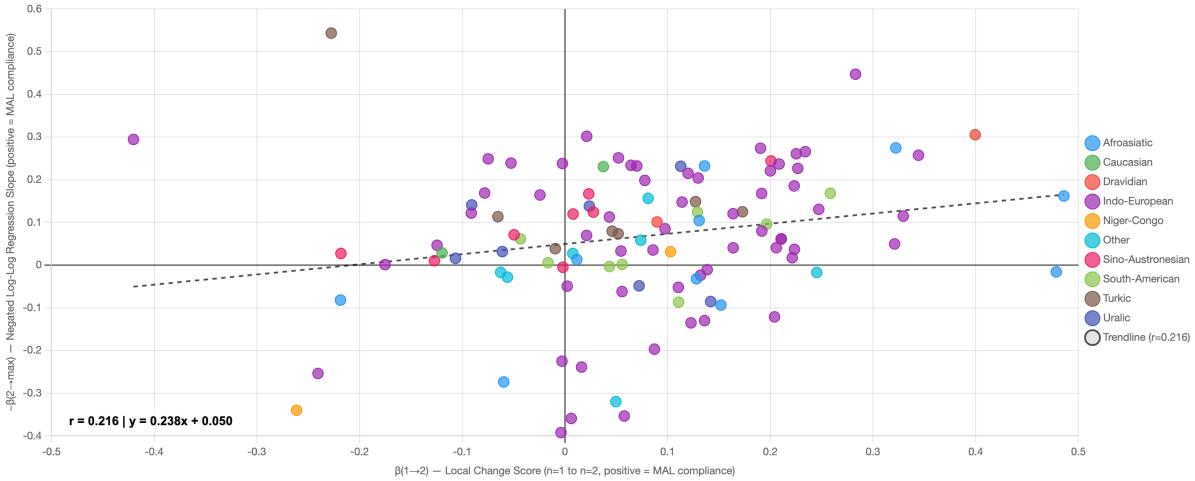

Figure 2: $\beta(1\rightarrow 2)$ vs $\beta(2\rightarrow \infty)$

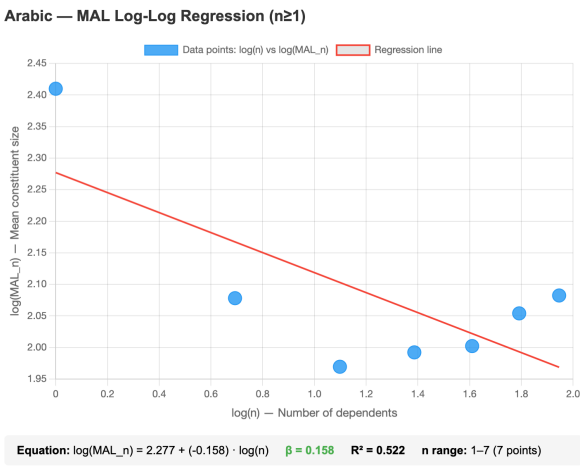

Figure 3: λMAL(Arabic).

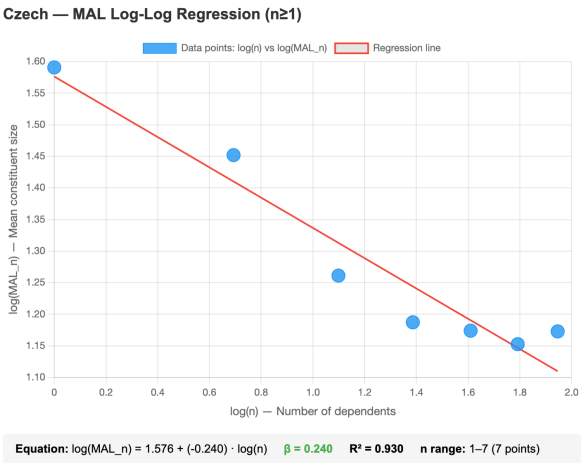

Figure 4: λMAL(Czech).

in the preverbal domain, against 7% in the postverbal domain, while 8% show a bilateral anti MAL

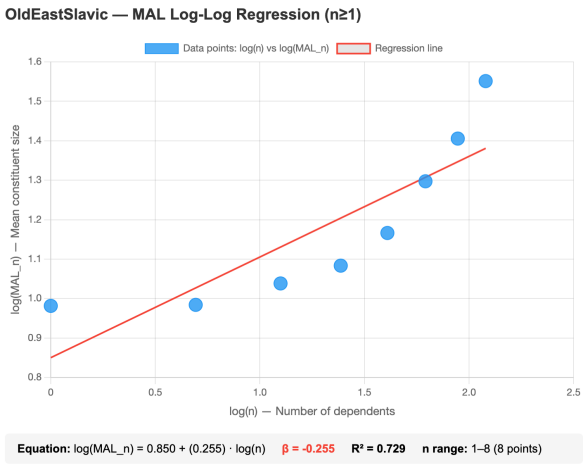

Figure 5: λMAL(OldEastSlavic).

Family MAL type	IE	Non-IE	Total
Anti Mal	6	4	10
Gray zone	22	37	59
MAL	38	24	62
Total	66	65	131

Figure 6: MAL per language families.

effect.

We can hence say that there is both a RMAL and an Anti LMAL bias - in other words, a ference for MAL in the postverbal domain and for Anti MAL in the preverbal domain. Importantly, in our data, like for the combined MAL, the three LMAL and RMAL types are also fairly well-distributed across language families (Table 5. However, the same is not true for language types, which is not surprising.

In the postverbal domain, we find that VO lan-

VO/OV MAL type	NDO	OV	VO	Total
Anti Mal	2	2	6	10
Gray zone	13	20	26	59
MAL	5	16	41	62
Total	20	38	73	131

Figure 7: MAL per VO/OV types.

Family RMAL type	IE	Non-IE	Total
Anti Mal	4	3	7
Gray zone	5	10	15
MAL	51	30	81
Total	60	43	103

Family LMAL type	IE	Non-IE	Total
Anti Mal	15	14	29
Gray zone	28	28	56
MAL	20	19	39
Total	63	61	124

Figure 8: RMAL and LMAL per language families.

VO/OV RMAL type	NDO	OV	VO	Total
Anti Mal	3	2	2	7
Gray zone	2	4	9	15
MAL	14	10	57	81
Total	19	16	68	103

VO/OV LMAL type	NDO	OV	VO	Total
Anti Mal	2	4	23	29
Gray zone	12	19	25	56
MAL	4	15	20	39
Total	18	38	68	124

Figure 9: RMAL and LMAL per VO/OV types.

guages show a more clear bias towards RMAL than OV language. More precisely, out of the 81 languages showing a RMAL effect, that is with a significant positive MAL effect in the postverbal domain, 57 (79%) are VO, against 10 (12%) OV languages. Also the anti RMAL preference is particularly rare in VO languages (2 out of 69 languages, 3%). Inversely, in the preverbal domain, we observe more OV languages with a LMAL preference than VO languages (51% vs. 38% of the total of 39 LMAL languages). Importantly, the anti LMAL preference is more frequent in VO languages than in OV languages (79% vs. 14% of the total of 29 anti LMAL languages). In other words, one the one hand, VO languages are significantly more likely to show a RMAL preference than OV languages do (84% vs. 63%), while they are also less likely to show an anti RMAL preference (3% vs. 13%). On the other hand, OV languages are more likely to show a LMAL preference than VO languages do (39% vs. 29%) and they are also less likely to show an anti LMAL preference (11% vs. 34%).

These differences are indeed not surprising, given that an asymmetry in terms of the length-based effects is indeed established in the literature between head-final and head-initial languages. They are also in line with previous observations on the difference between the effect of phrasal length in pre- and postverbal domains (Faghiri and Samvelian, 2020).

5.1. Zooming into different types

It would not possible to present the results for all the languages with enough data. An online website will all the results will be proposed with the final version. Here we will present on some remarkable indvidual languages in particular among the those which contradic MAL in one or both verbal domains.

Starting with the preverbal domain, we find 29 Anti LMAL languages, among which 13 are also RMAL. Otherwise, they are mostly in the grey zone when the bilateral MAL is considered. Two of these languages are remarkable: Occitan (Romance, France) is MAL, while Naija (English pidgincreole, Nigeria) is Anti MAL. Naija is also remarkable, since it is the most strongly Anti LMAL language of our sample, following a straight power law (Figures 10 and 11).

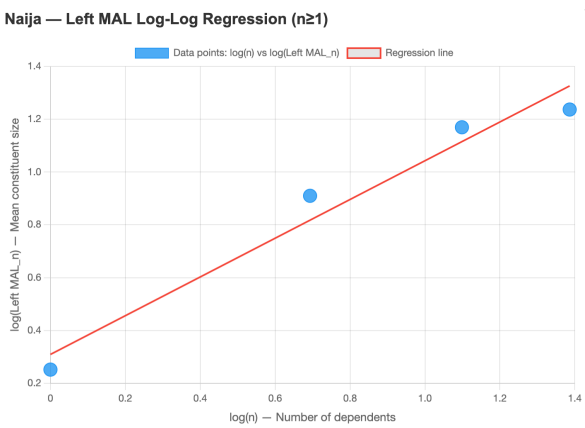

Figure 10: λMAL(Naija).

As for the postverdal domain, we find the follwoing 7 Anti RMAL languages: Khoekhoe, Bambara, Egyptian, OldEastSlavic, WestArmenian, Gothic,

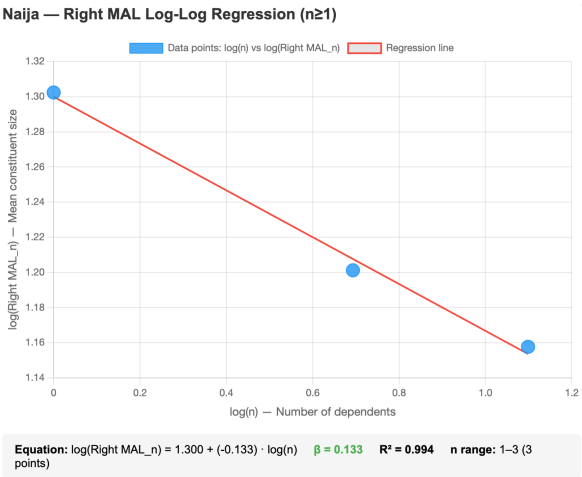

Figure 11: λMAL(Naija).

and Latin. It is worth noting that among them, 4 are corpora of ancient languages: Egyptian, OldEastSlavic, Gothic, and Latin. There are only 3 languages which are both Anti LMAL and Anti RMAL: Bambara, OldEastSlavic, and Latin.

6. Conclusion

7. Acknowledgements

A. MAL results for all languages

Table 1: MAL, LMAL and RMAL mini log-log plots and $\beta(1 \to \infty)$ for all languages. Green = MAL ($\beta > 0.1$), red = Anti-MAL ($\beta < -0.1$), orange = grey zone.

Language	Family	Type	MAL	β	LMAL	β	RMAL	β
Abkhaz	Caucasia	OV		-0.07		+0.07	—	—
Afrikaans	IE	OV		+0.18		+0.19		+0.55
Akkadian	Afroasia	OV		+0.05		+0.09	—	—
Albanian	IE	VO		-0.05		+0.12		+0.47
Amharic	Afroasia	OV		+0.07		+0.30		-0.00
AncientGreek	IE	mix		+0.05		-0.06		+0.01
AncientHebrew	Afroasia	VO		+0.03		-0.13		+0.03
Arabic	Afroasia	VO		+0.16		-0.12		+0.10
Arhuaco	South-Am	OV		+0.13		+0.25	—	—
Armenian	IE	mix		+0.06		-0.05		+0.15
Bambara	Nig-Co.	OV		-0.29		-0.51		-0.20
Basque	Other	OV		+0.06		+0.18		+0.11
Bavarian	IE	mix		+0.13		+0.09		+0.75

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Language	Family	Type	MAL	β	LMAL	β	RMAL	β
Beja	Afroasia	OV		+0.38		+0.40	—	—
Belarusian	IE	VO		+0.18		+0.09		+0.34
Bhojpuri	IE	OV		-0.04		+0.09	—	—
Borôro	South-Am	OV		+0.13		+0.23		+0.10
Breton	IE	VO		+0.12	—	—		+0.20
Buglere	South-Am	OV		-0.01		-0.05	—	—
Bulgarian	IE	VO		+0.17		+0.17		+0.33
Buryat	Other	OV		+0.16		+0.19	—	—
Cantonese	Sino-Aus	VO		-0.05		-0.24		+0.39
CappadocianGreek	IE	VO	—	—	—	—		+0.24
Catalan	IE	VO		+0.26		+0.25		+0.36
Chhintange	Sino-Aus	OV		+0.21		+0.19	—	—
Chinese	Sino-Aus	VO		-0.04		-0.15		+0.53
Chukot	Other	mix		-0.05		+0.01	—	—
ClassArmenian	IE	mix		+0.00		-0.07		+0.01
ClassChinese	Sino-Aus	VO		+0.07		+0.04		+0.01
Coptic	Afroasia	VO		-0.15		-0.49		-0.03
Croatian	IE	VO		+0.27		+0.15		+0.35
Czech	IE	VO		+0.24		+0.22		+0.42
Danish	IE	VO		+0.13		+0.26		+0.31
Dutch	IE	mix		+0.09		+0.15		+0.49
Egyptian	Afroasia	VO		+0.08	—	—		-0.20
English	IE	VO		+0.06		-0.30		+0.19
Erzya	Uralic	mix		-0.02		-0.00		+0.27
Estonian	Uralic	VO		+0.21		+0.18		+0.45
Faroese	IE	VO		-0.08		-0.01		+0.02
Finnish	Uralic	VO		+0.11		+0.06		+0.38
French	IE	VO		+0.25		+0.01		+0.32
FrisianDutch	IE	mix		+0.33	—	—	—	—
Gaelic	IE	VO		+0.35	—	—		+0.24
Galician	IE	VO		+0.39		+0.24		+0.50
Gbaya	Nig-Co.	VO	—	—	—	—		+0.17
Georgian	Caucasia	mix		+0.15		-0.04		+0.44
German	IE	mix		+0.17		+0.20		+0.54
Gheg	IE	VO		+0.11		-0.04		+0.31
Gothic	IE	VO		-0.12		-0.07		-0.12
Greek	IE	VO		+0.21		-0.04		+0.32
Guajajára	South-Am	VO		+0.08	—	—		+0.18
Haitian	IE	VO		+0.19		-0.29		+0.30

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Language	Family	Type	MAL	β	LMAL	β	RMAL	β
Hausa	Afroasia	VO		-0.16		-0.40		+0.07
Hebrew	Afroasia	VO		+0.19		+0.20		+0.32
HighlandNahuatl	South-Am	VO		+0.23		+0.10		+0.43
Hindi	IE	OV		+0.10		+0.03		+1.03
Hungarian	Uralic	mix		+0.05		+0.08		+0.24
Icelandic	IE	VO		-0.03		-0.17		+0.22
Indonesian	Sino-Aus	VO		+0.11		-0.07		+0.31
Irish	IE	VO		+0.05		-0.31		+0.15
Italian	IE	VO		+0.15		+0.00		+0.22
Japanese	Other	OV		+0.03		+0.02	—	—
Javanese	Sino-Aus	VO		+0.22		-0.00		+0.21
Karo	South-Am	OV		+0.05		+0.07	—	—
Kazakh	Turkic	OV		+0.06		+0.07	—	—
Khoekhoe	Other	OV		-0.04		+0.07		-0.37
Kiche	South-Am	VO		+0.05	—	—		+0.21
Komi	Uralic	mix		-0.04		+0.01		+0.20
Korean	Other	OV		+0.13		+0.12		-0.06
Kurmanji	IE	OV		+0.14		-0.13	—	—
Kyrgyz	Turkic	OV		+0.15		+0.15	—	—
Latin	IE	mix		-0.23		-0.14		-0.10
Latvian	IE	VO		+0.11		-0.02		+0.30
Ligurian	IE	VO		-0.07		+0.05		+0.55
Lithuanian	IE	VO		+0.15		-0.08		+0.38
LowSaxon	IE	mix		+0.10		+0.14		+0.51
Maghrebi	Afroasia	VO		+0.12		+0.15		+0.17
Maltese	Afroasia	VO		+0.30		-0.07		+0.41
Manx	IE	VO	—	—	—	—		+0.39
Marathi	IE	OV		+0.12		+0.10	—	—
MbyáGuaraní	South-Am	mix		-0.01		+0.00	—	—
MiddleFrench	IE	VO		+0.12		-0.29		+0.15
Moksha	Uralic	VO		+0.07		+0.06	—	—
Naija	IE	VO		-0.25		-0.73		+0.13
Nhengatu	South-Am	VO		+0.03		+0.14		+0.10
NorthSami	Uralic	VO		+0.03		-0.00		+0.19
Norwegian	IE	VO		+0.22		+0.18		+0.39
Occitan	IE	VO		+0.21		-0.15		+0.33
OldChurchSlavonic	IE	VO		-0.13		-0.17		-0.08
OldEastSlavic	IE	mix		-0.26		-0.21		-0.16
OldFrench	IE	VO		-0.05		-0.18		+0.15

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Language	Family	Type	MAL	β	LMAL	β	RMAL	β
OldProvençal	IE	VO		-0.20		-0.44		-0.01
OttomanTurkish	Turkic	OV		+0.06		+0.03	—	—
Persian	IE	OV		-0.01		-0.09		+0.68
Pesh	Other	OV		+0.10		-0.04	—	—
Polish	IE	VO		+0.14		+0.12		+0.26
Pomak	IE	VO		+0.05		+0.10		+0.19
Portuguese	IE	VO		+0.13		+0.13		+0.17
Romanian	IE	VO		+0.09		-0.11		+0.21
Russian	IE	VO		+0.11		+0.07		+0.32
Sanskrit	IE	OV		-0.02		+0.03		+0.42
Serbian	IE	VO		+0.17		+0.23		+0.37
Sicilian	IE	VO		+0.32		-0.08		+0.36
Sindhi	IE	OV		-0.05		+0.05		+0.84
SkoltSami	Uralic	mix		+0.03		-0.10	—	—
Slovak	IE	VO		+0.14		+0.24		+0.34
Slovenian	IE	VO		+0.23		+0.06		+0.42
Spanish	IE	VO		+0.21		+0.22		+0.26
Swedish	IE	VO		+0.11		+0.10		+0.35
SwissGerman	IE	OV		-0.00		+0.17		+1.01
Tamil	Dravid.	OV		+0.35		+0.38	—	—
Tangkhul	Sino-Aus	OV		+0.11		+0.09	—	—
Tatar	Turkic	OV		-0.08		-0.13	—	—
Teko	South-Am	mix		+0.10		+0.13	—	—
Telugu	Dravid.	OV		+0.09		+0.09	—	—
Thai	Sino-Aus	VO		+0.09		-0.22		+0.30
Turkish	Turkic	OV		+0.14		+0.14		-0.06
TurkishGerman	Turkic	OV		+0.05		+0.13		+0.40
Ukrainian	IE	VO		+0.19		-0.04		+0.32
UpperSorbian	IE	VO		+0.50		+0.50		+0.72
Urdu	IE	OV		+0.14		+0.05		+0.76
Uyghur	Turkic	OV		+0.02		+0.04	—	—
Uzbek	Turkic	OV		+0.14		+0.16	—	—
Vietnamese	Sino-Aus	VO		+0.01		-0.20		+0.24
Welsh	IE	VO		+0.63		-0.02		+0.70
WestArmenian	IE	mix		-0.03		-0.10		+0.17
WestNahuatl	South-Am	VO		+0.04		+0.03		-0.13
Wolof	Nig-Co.	VO		+0.06		-0.32		+0.22
Wu	Sino-Aus	VO		-0.00		-0.05	—	—
Xibe	Other	OV		-0.13		-0.13	—	—

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Language	Family	Type	MAL	β	LMAL	β	RMAL	β
Yiddish	IE	VO		+0.04		-0.10		+0.29
Yoruba	Nig-Co.	VO		+0.34		-0.21		+0.49
Yupik	Other	mix		-0.05	—	—	—	—
Zaar	Afroasia	VO		+0.01		-0.28		+0.35

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